

INSERT INTO Employees (EmpID, EmpName, DepartmentID, Salary, ManagerID) VALUES

(1, 'Alice', 101, 70000, 5),
(2, 'Bob', 102, 60000, 5),
(3, 'Charlie', 103, 80000, 6),
(4, 'David', 101, 75000, 6),
(5, 'Emma', 102, 90000, NULL),
(6, 'Frank', 103, 85000, NULL);

INSERT INTO Departments (DeptID, DeptName, LocationID) VALUES

(101, 'HR', 1),
(102, 'IT', 2),
(103, 'Finance', 3);

INSERT INTO Locations (LocationID, City, Country) VALUES

(1, 'New York', 'USA'),
(2, 'Los Angeles', 'USA'),
(3, 'London', 'UK');

INSERT INTO Projects (ProjectID, ProjectName, EmpID) VALUES

(1, 'Project A', 1),
(2, 'Project B', 2),

(3, 'Project C', 3),

(4, 'Project D', 4),

(5, 'Project E', 5),

(6, 'Project F', 6);

1. Retrieve all employees along with their department names.

```
SELECT e.EmpName, d.DeptName
FROM Employees e
INNER JOIN Departments d ON e.DepartmentID = d.DeptID;
```

2. Get all employees and their corresponding project names.

```
SELECT e.EmpName, p.ProjectName
FROM Employees e
LEFT JOIN Projects p ON e.EmpID = p.EmpID;
```

3. Find employees who work in the 'IT' department.

```
SELECT e.EmpName
FROM Employees e
INNER JOIN Departments d ON e.DepartmentID = d.DeptID
WHERE d.DeptName = 'IT';
```

4. List employees along with their manager names.

```
SELECT e.EmpName AS Employee, m.EmpName AS Manager
FROM Employees e
LEFT JOIN Employees m ON e.ManagerID = m.EmpID;
```

5. Get all departments and their associated employees (including those without employees).

```
SELECT d.DeptName, e.EmpName
FROM Departments d
LEFT JOIN Employees e ON d.DeptID = e.DepartmentID;
```

6. Retrieve all employees and their city and country based on their department's location.

```
SELECT e.EmpName, l.City, l.Country
FROM Employees e
INNER JOIN Departments d ON e.DepartmentID = d.DeptID
INNER JOIN Locations l ON d.LocationID = l.LocationID;
```

7. Find employees who have not been assigned to any project.

```
SELECT e.EmpName
FROM Employees e
LEFT JOIN Projects p ON e.EmpID = p.EmpID
WHERE p.ProjectID IS NULL;
```

8. Show all projects with the employees working on them.

```
SELECT p.ProjectName, e.EmpName
FROM Projects p
LEFT JOIN Employees e ON p.EmpID = e.EmpID;
```

9. List employees who work in 'New York'.

```
SELECT e.EmpName
FROM Employees e
INNER JOIN Departments d ON e.DepartmentID = d.DeptID
INNER JOIN Locations l ON d.LocationID = l.LocationID
WHERE l.City = 'New York';
```

10. Retrieve employees and their salaries along with department names.

```
SELECT e.EmpName, e.Salary, d.DeptName
```

```
FROM Employees e
INNER JOIN Departments d ON e.DepartmentID = d.DeptID;
```

11. Get the total number of employees in each department.

```
SELECT d.DeptName, COUNT(e.EmpID) AS TotalEmployees
FROM Departments d
LEFT JOIN Employees e ON d.DeptID = e.DepartmentID
GROUP BY d.DeptName;
```

12. Find the highest-paid employee in each department.

```
SELECT d.DeptName, e.EmpName, e.Salary
FROM Employees e
INNER JOIN Departments d ON e.DepartmentID = d.DeptID
WHERE e.Salary = (SELECT MAX(Salary) FROM Employees WHERE
DepartmentID = e.DepartmentID);
```

13. Retrieve employees who work in locations outside the USA.

```
SELECT e.EmpName
FROM Employees e
INNER JOIN Departments d ON e.DepartmentID = d.DeptID
INNER JOIN Locations l ON d.LocationID = l.LocationID
WHERE l.Country <> 'USA';
```

14. Find employees who are also managers.

```
SELECT DISTINCT m.EmpName AS Manager
FROM Employees e
INNER JOIN Employees m ON e.ManagerID = m.EmpID;
```

15. Get a list of all departments that have employees.

```
SELECT DISTINCT d.DeptName
FROM Departments d
INNER JOIN Employees e ON d.DeptID = e.DepartmentID;
```

16. Retrieve the total salary expenditure per department.

```
SELECT d.DeptName, SUM(e.Salary) AS TotalSalary
FROM Departments d
LEFT JOIN Employees e ON d.DeptID = e.DepartmentID
GROUP BY d.DeptName;
```

17. List employees who earn more than their manager.

```
SELECT e.EmpName, e.Salary
FROM Employees e
INNER JOIN Employees m ON e.ManagerID = m.EmpID
WHERE e.Salary > m.Salary;
```

18. Show the average salary per department.

```
SELECT d.DeptName, AVG(e.Salary) AS AvgSalary
FROM Departments d
LEFT JOIN Employees e ON d.DeptID = e.DepartmentID
GROUP BY d.DeptName;
```

19. Find employees who do not have a manager.

```
SELECT EmpName
FROM Employees
WHERE ManagerID IS NULL;
```

20. Retrieve employees working on multiple projects.

```
SELECT e.EmpName
FROM Employees e
INNER JOIN Projects p ON e.EmpID = p.EmpID
GROUP BY e.EmpName
HAVING COUNT(p.ProjectID) > 1;
```

21. Find the city with the highest number of employees.

```
SELECT l.City, COUNT(e.EmpID) AS EmployeeCount
FROM Employees e
INNER JOIN Departments d ON e.DepartmentID = d.DeptID
INNER JOIN Locations l ON d.LocationID = l.LocationID
GROUP BY l.City
ORDER BY EmployeeCount DESC
LIMIT 1;
```

22. Retrieve employees and their project names, even if they are not assigned any project.

```
SELECT e.EmpName, p.ProjectName
FROM Employees e
LEFT JOIN Projects p ON e.EmpID = p.EmpID;
```

23. Get all departments, including those without employees.

```
SELECT d.DeptName
FROM Departments d
LEFT JOIN Employees e ON d.DeptID = e.DepartmentID;
```

24. Show locations where no departments exist.

```
SELECT l.City
FROM Locations l
LEFT JOIN Departments d ON l.LocationID = d.LocationID
WHERE d.DeptID IS NULL;
```

25. List employees who work on more than one project.

```
SELECT e.EmpName
FROM Employees e
INNER JOIN Projects p ON e.EmpID = p.EmpID
GROUP BY e.EmpName
HAVING COUNT(p.ProjectID) > 1;
```

26. Find departments that do not have employees.

```
SELECT d.DeptName
FROM Departments d
LEFT JOIN Employees e ON d.DeptID = e.DepartmentID
WHERE e.EmpID IS NULL;
```

27. Retrieve projects assigned to employees with salaries above 80,000.

```
SELECT p.ProjectName, e.EmpName
FROM Projects p
INNER JOIN Employees e ON p.EmpID = e.EmpID
WHERE e.Salary > 80000;
```

28. Find departments where the total salary is more than 150,000.

```
SELECT d.DeptName, SUM(e.Salary) AS TotalSalary
FROM Departments d
INNER JOIN Employees e ON d.DeptID = e.DepartmentID
GROUP BY d.DeptName
HAVING SUM(e.Salary) > 150000;
```

29. List managers along with the number of employees they manage.

```
SELECT m.EmpName AS Manager, COUNT(e.EmpID) AS TotalEmployees
FROM Employees m
INNER JOIN Employees e ON m.EmpID = e.ManagerID
GROUP BY m.EmpName;
```

30. Retrieve employees whose department is in 'London' but who do not work on any project.

```
SELECT e.EmpName
FROM Employees e
INNER JOIN Departments d ON e.DepartmentID = d.DeptID
INNER JOIN Locations l ON d.LocationID = l.LocationID
LEFT JOIN Projects p ON e.EmpID = p.EmpID
WHERE l.City = 'London' AND p.ProjectID IS NULL;
```

